

Makilesh M

Machine Learning Engineer - Multimodal AI, RAG, Production Pipelines

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SUMMARY

Oracle-certified AI engineer with hands-on experience in PyTorch/TensorFlow, computer vision, RAG, data pipelines, and production AI systems. Strong foundation for multimodal AI roles involving Python, model evaluation, data ingestion, inference optimization, and deployment.

RELEVANT SKILLS

ML & Vision: PyTorch, TensorFlow, OpenCV, CNNs, Transformers, Deep Learning, uncertainty estimation, model calibration

AI Systems: RAG, LangGraph, LangChain, Milvus, Pinecone, FAISS, FastAPI, Docker, Hugging Face, Python, Pandas, NumPy

EXPERIENCE

AI Developer Intern | Shamla Tech Solutions | Sep 2025 - Mar 2026

- Built AI data and automation pipelines with async scraping, semantic filtering, batch processing, and downstream LLM workflows.
- Delivered latency-sensitive voice AI systems with async queues, memory-aware concurrency, and fallback design.

SELECTED PROJECTS

CUA-DELC Lung Disease Classifier

- Built dynamic ensemble using InceptionV3, MobileNetV2, and VGG19 for four-class lung disease classification.
- Implemented confidence and uncertainty-aware weighting, temperature-scaled calibration, and entropy-based uncertainty estimation.

Agentic RAG System | github.com/Makilesh/AGENTIC_RAG

- Built RAG system supporting PDF, DOCX, PPTX, Excel, and TXT ingestion with hybrid Milvus retrieval and FastAPI deployment.
- Implemented self-corrective retrieval loop and ~200ms retrieval latency.

AI-Powered Voice Agent | github.com/Makilesh/voice_engine_MVP

- Built production-grade real-time voice agent with 200-500ms latency, <150ms barge-in detection, and async STT/LLM/TTS pipeline.

EDUCATION & CERTIFICATIONS

B.E. Computer Science Engineering, Sri Krishna College of Engineering and Technology | CGPA: 8.0 | Expected May 2026

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional; Oracle Cloud Infrastructure 2025 Certified Data Science Professional; Udemy Complete ML, NLP Bootcamp - MLOps & Deployment.